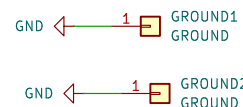


[illegible]

Front Circuits

The RP2040 GPIO pins can support a max of 50mA TOTAL. RP2040 runs at 3.3v

Each pin should not draw more than 16mA
 $3.3 / 0.016 = 206$

Let's go higher and say 220R

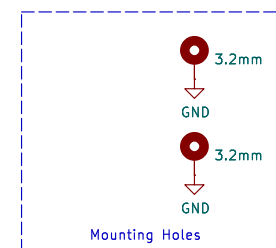
The image contains two circuit diagrams labeled "Left Front" and "Right Front". Both are powered by Vbatt and have a common ground.

Left Front Circuit:

- Vbatt is connected to a 10R resistor (R7).
- A signal from GPIO_FRONT passes through a 220R resistor (R2) to the base of a BC849 NPN transistor (TR2).
- The emitter of TR2 is grounded.
- The collector of TR2 is connected to the gate of a PMV45EN2 MOSFET (FET2).
- The drain of FET2 is connected to Vbatt through a 10K resistor (R8) and a 10uF capacitor (C2) to ground.
- Two LEDs, FLED1 and FLED2 (TLCR5800), are connected between Vbatt and the drain of FET2.

Right Front Circuit:

- Vbatt is connected to a 220R resistor (R12) leading to Vcc.
- A signal from GPIO_FRONT passes through a 10K resistor (R9) to the base of a BC849 NPN transistor (TR2).
- The emitter of TR2 is grounded.
- The collector of TR2 is connected to the gate of a PMV45EN2 MOSFET (FET2).
- The drain of FET2 is connected to Vbatt through a 10K resistor (R11) and a 10uF capacitor (C3) to ground.
- Two photodiodes, LFPD1 and LFPD2 (BPW96B), are connected between Vbatt and the drain of FET2.
- The output of LFPD1 is connected to LEFT_FRONT_TP.
- The output of LFPD2 is connected to RIGHT_FRONT_TP.



Id: 1/1